

Neighborhood Poverty vs. Family Income as Predictors of Academic Outcomes

2026-04-22

I consulted Claude.io for information about how to write this code. Lines 254-283 of this code file were copied from Claude to create the interaction plots. I also used it for 191 to 211 to combine the posterior graphs of

<https://stackoverflow.com/questions/bayesian-inference-in-R> (<https://stackoverflow.com/questions/bayesian-inference-in-R>) was consulted for information (Be sure to include enough information that I can find the page you looked at!).

Research Question: Neighborhood Poverty vs. Family Income as Predictors of Academic Outcomes

Data Ingestion

```
library(tidyverse)
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

```
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.1.5
## ✓ forcats    1.0.0      ✓ stringr    1.5.1
## ✓ ggplot2    4.0.2      ✓ tibble     3.2.1
## ✓ lubridate  1.9.3      ✓ tidyr      1.3.1
## ✓ purrr      1.0.2
## — Conflicts — tidyverse_conflicts() —
## X dplyr::filter() masks stats::filter()
## X dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
test_scores<- read_csv("seda_geodist_pool_cs_6.0.csv")
```

```
## Rows: 144246 Columns: 42
## — Column specification —————
## Delimiter: ","
## chr (4): sedaleaname, stateabb, subcat, subgroup
## dbl (38): sedalea, fips, multi_comp, gap, gradecenter, tot_asmts, cellcount,...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
socioeconomics<- read_csv("seda_cov_geodist_pool_6.0.csv")
```

```
## Rows: 12713 Columns: 98
## — Column specification —————
## Delimiter: ","
## chr (4): sedaleaname, stateabb, gslo, urbanicity
## dbl (94): sedalea, fips, gshi, urban, suburb, town, rural, locale_city_large...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
#Merge using an inner join.
```

```
merged<- inner_join(test_scores, socioeconomics, by = "sedalea")
```

```
#Subsetting dataset to use the variables i need
#For all students
```

```
df_all <- merged %>%
  filter(subgroup == "all") %>%
  select(sedalea, sedaleaname.x, cs_mn_avg_ol,
         lninc50avgall, povertyavgall, urbanicity) %>%
  rename(distname      = sedaleaname.x,
         test_score    = cs_mn_avg_ol,
         fam_income    = lninc50avgall,
         nbhd_poverty  = povertyavgall) %>%
  drop_na(test_score, fam_income, nbhd_poverty, urbanicity) %>%
  mutate(urbanicity = factor(urbanicity,
                             levels = c("City", "Suburb", "Town", "Rural")))
```

```
table(df_all$urbanicity)
```

```
##
##   City Suburb   Town   Rural
##   687   2809   2298   6541
```

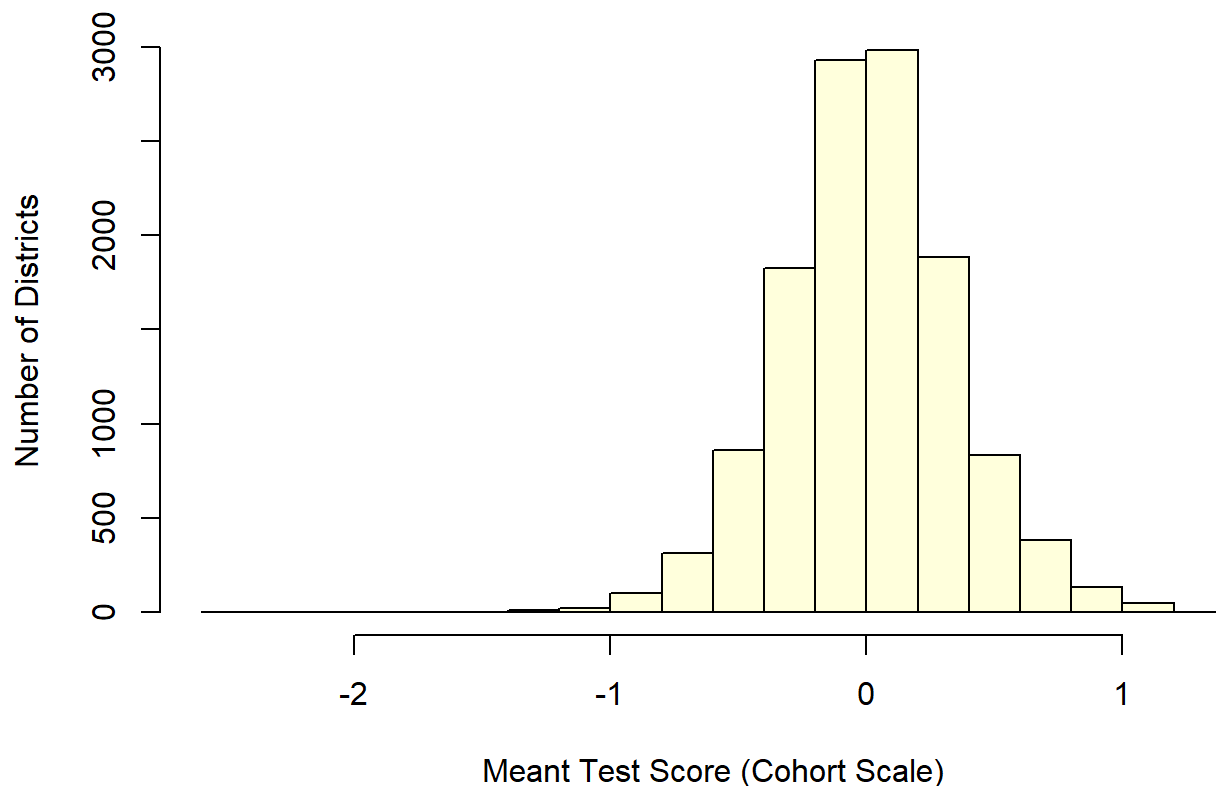
```
summary(df_all)
```

```
##      sedalea      distname      test_score      fam_income
## Min.   : 100005 Length:12335 Min.    :-2.437870 Min.    : 9.838
## 1st Qu.:1810095 Class :character 1st Qu.: -0.204875 1st Qu.:10.645
## Median :3025530 Mode  :character Median : 0.006946 Median :10.819
## Mean   :3004977      Mean   : 0.007750 Mean   :10.859
## 3rd Qu.:4100022      3rd Qu.: 0.215358 3rd Qu.:11.032
## Max.    :5606240      Max.    : 1.330566 Max.    :12.334
## nbhd_poverty      urbanicity
## Min.    :0.007008 City   : 687
## 1st Qu.:0.083713 Suburb:2809
## Median :0.121814 Town   :2298
## Mean    :0.129942 Rural  :6541
## 3rd Qu.:0.165150
## Max.    :0.471623
```

Univariate Analysis for All students

```
library(ggplot2)
#Test Scores
hist(df_all$test_score, main = paste("Distribution of Average Test Scores by District"), xlab =
"Meant Test Score (Cohort Scale)", ylab = "Number of Districts", col = "lightyellow")
```

Distribution of Average Test Scores by District



```
#Family Income
```

```
hist(df_all$fam_income, main = paste("Distiribution of Median Household Income by District"), xlab = "Log Median Family Income Averaged over all Students", ylabcol = "lightblue2")
```

```
## Warning in plot.window(xlim, ylim, "", ...): "ylabcol" is not a graphical parameter
```

```
## Warning in title(main = main, sub = sub, xlab = xlab, ylab = ylab, ...):  
## "ylabcol" is not a graphical parameter
```

```
## Warning in axis(1, ...): "ylabcol" is not a graphical parameter
```

```
## Warning in axis(2, at = yt, ...): "ylabcol" is not a graphical parameter
```

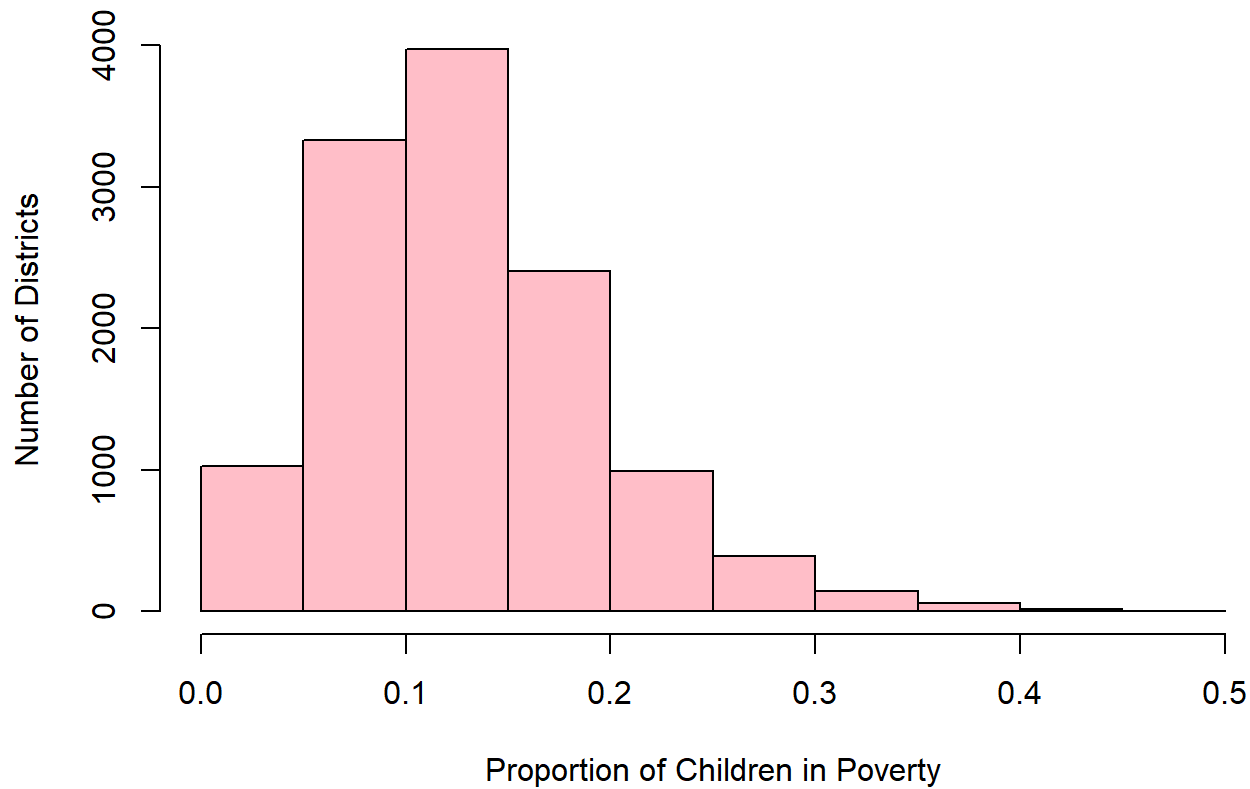
Distiribution of Median Household Income by District



```
#Neighborhood Poverty Rate
```

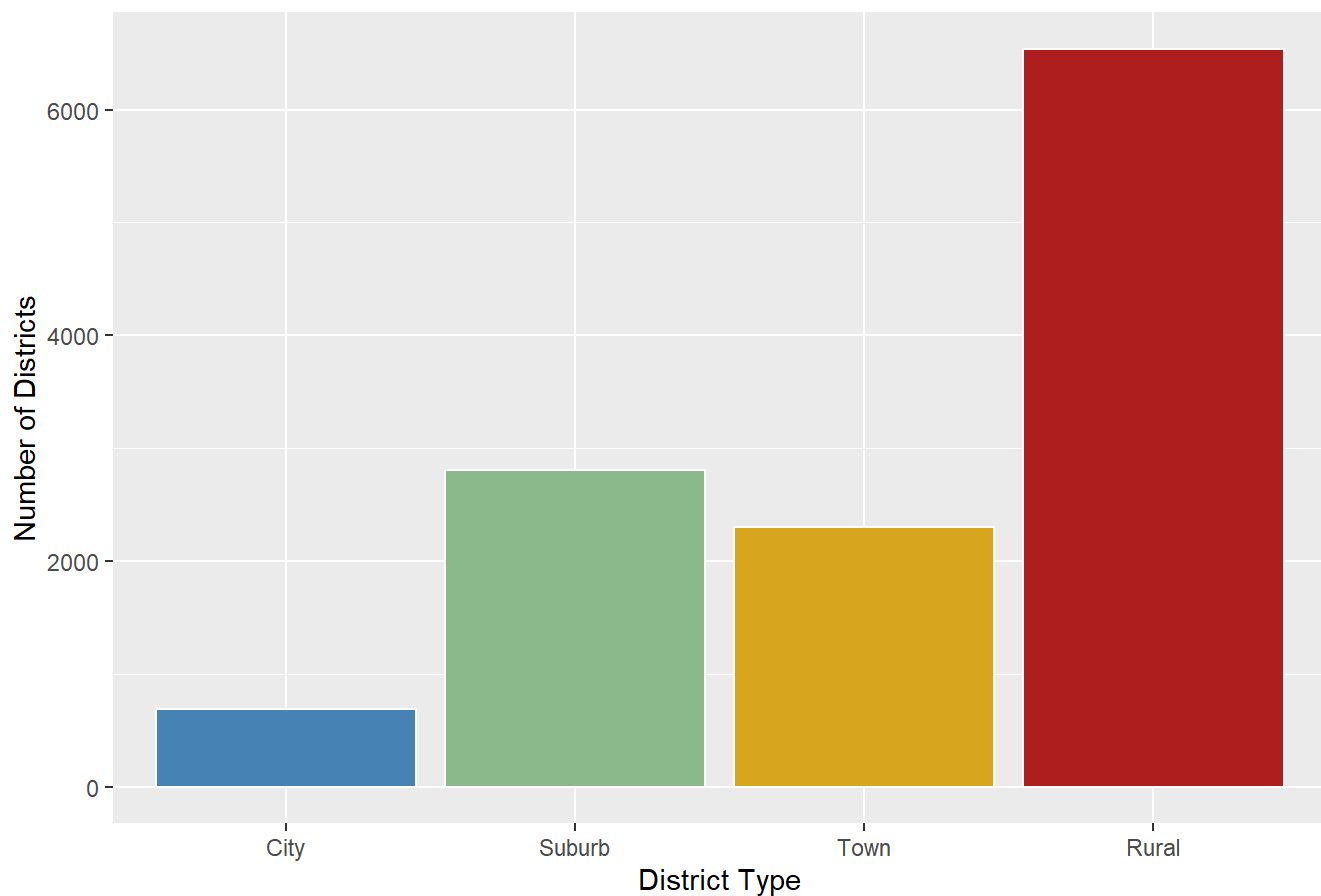
```
hist(df_all$nbhd_poverty, main = paste("Distribution of Child Poverty over Districts"), xlab = "Proportion of Children in Poverty", ylab = "Number of Districts", col="pink")
```

Distribution of Child Poverty over Districts



```
#Test scores over Neighborhood type
ggplot(df_all, aes(x = urbanicity, fill = urbanicity)) +
  geom_bar(color = "white") +
  scale_fill_manual(values = c("City" = "steelblue",
                              "Suburb" = "darkseagreen",
                              "Town" = "goldenrod",
                              "Rural" = "firebrick")) +
  labs(title = "Number of Districts by Urbanicity",
       x = "District Type",
       y = "Number of Districts") +
  theme(legend.position = "none")
```

Number of Districts by Urbanicity



Bivariate Analysis All Students

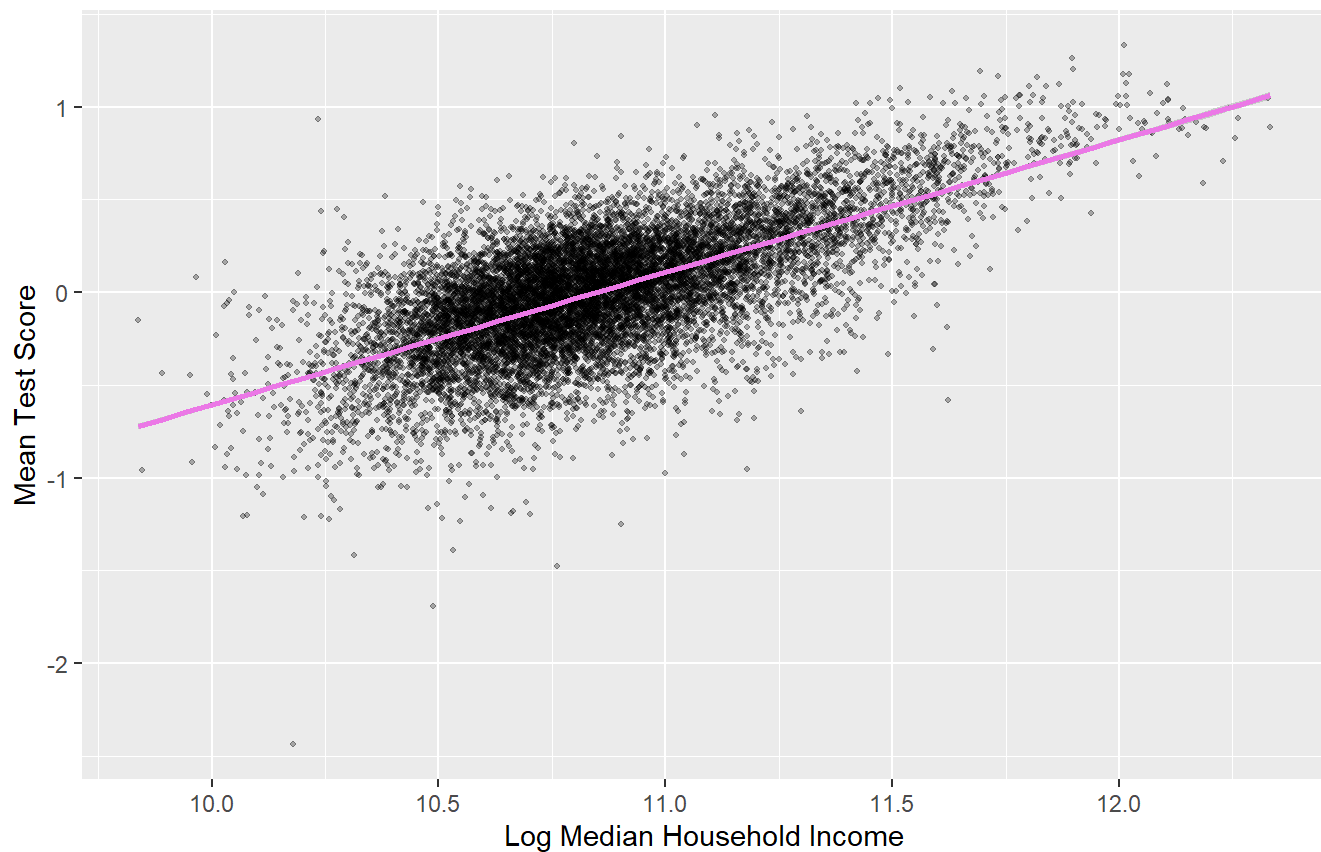
```
library(ggplot2)

#Family Income vs Test Scores
ggplot(df_all, aes(x = fam_income, y = test_score)) +
  geom_point(alpha = 0.3, size = 0.8) +
  geom_smooth(method = "lm", color = "orchid2", se = TRUE) +
  labs(title = "Family Income vs. Test Scores",
       subtitle = "The conventional story",
       x = "Log Median Household Income",
       y = "Mean Test Score")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

Family Income vs. Test Scores

The conventional story

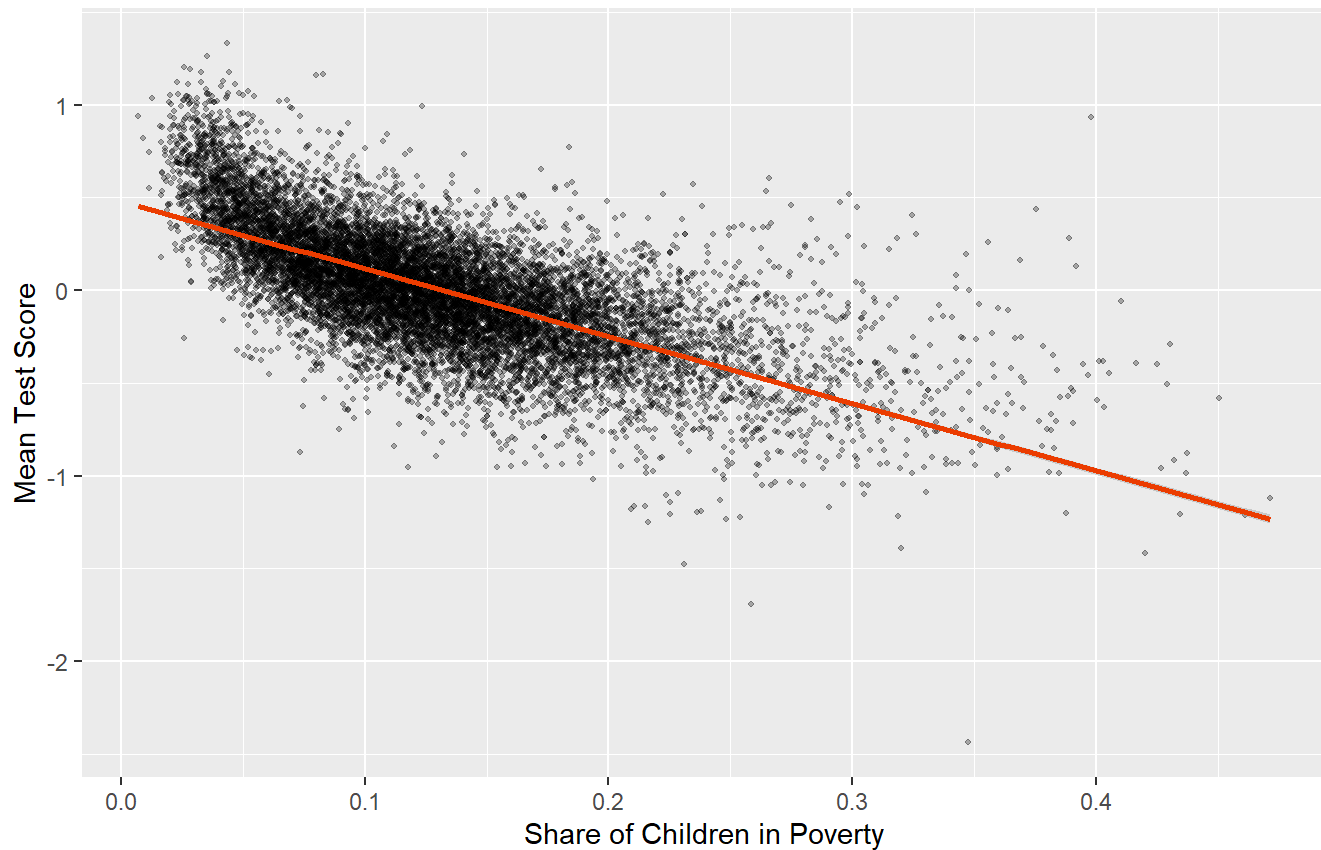


```
# Neighborhood Poverty vs Test Scores
ggplot(df_all, aes(x = nbhd_poverty, y = test_score)) +
  geom_point(alpha = 0.3, size = 0.8) +
  geom_smooth(method = "lm", color = "orangered2", se = TRUE) +
  labs(title = "Neighborhood Poverty Rate vs. Test Scores",
        subtitle = "The neighborhood story",
        x = "Share of Children in Poverty",
        y = "Mean Test Score")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

Neighborhood Poverty Rate vs. Test Scores

The neighborhood story

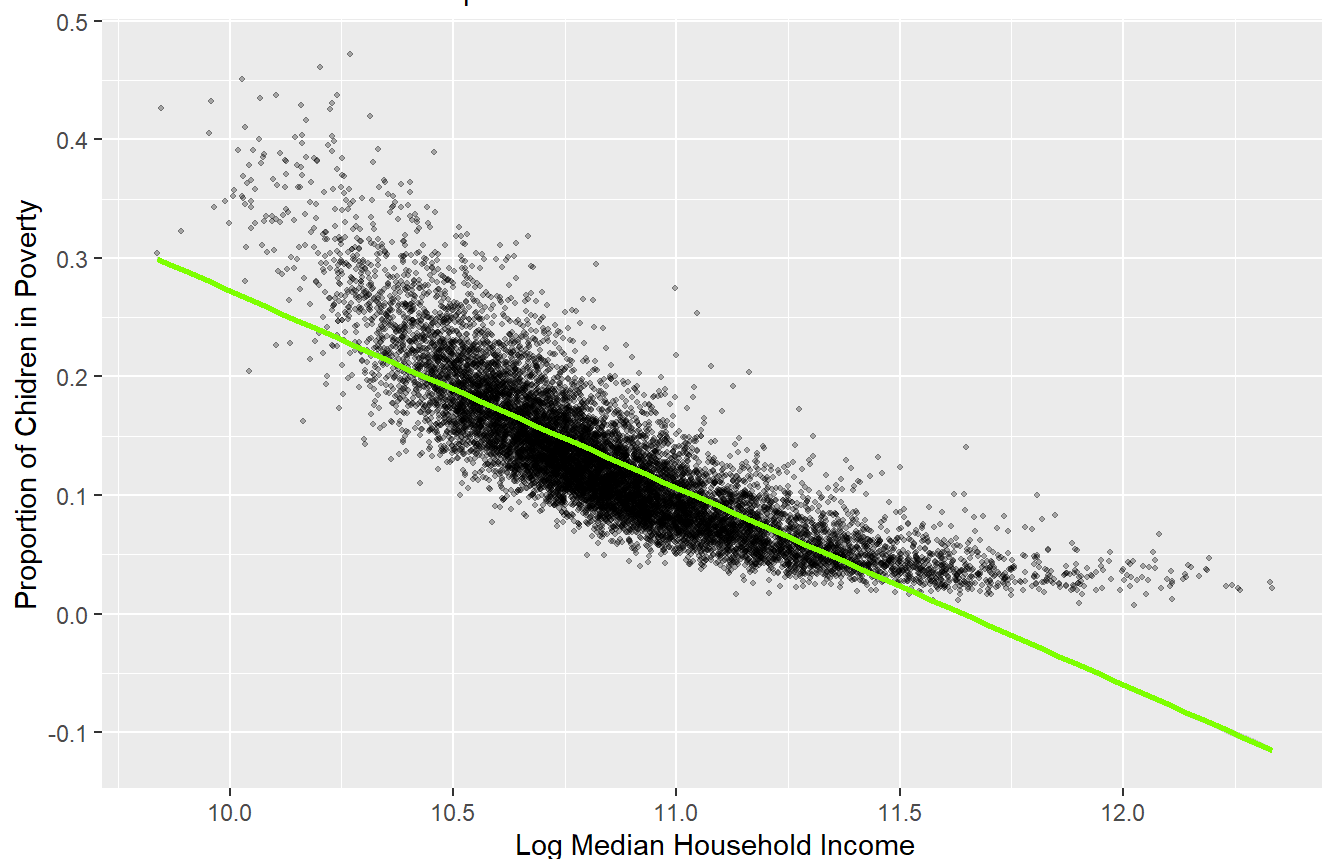


```
#checking for collinearity
#Family Income vs Neighborhood Poverty
ggplot(df_all, aes(x = fam_income, y = nbhd_poverty)) +
  geom_point(alpha = 0.3, size = 0.8) +
  geom_smooth(method = "lm", color = "chartreuse", se = TRUE) +
  labs(title = "Family Income vs Neighborhood Poverty",
        subtitle = "How Correlated are the two predictors",
        x = "Log Median Household Income",
        y = "Proportion of Children in Poverty")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

Family Income vs Neighborhood Poverty

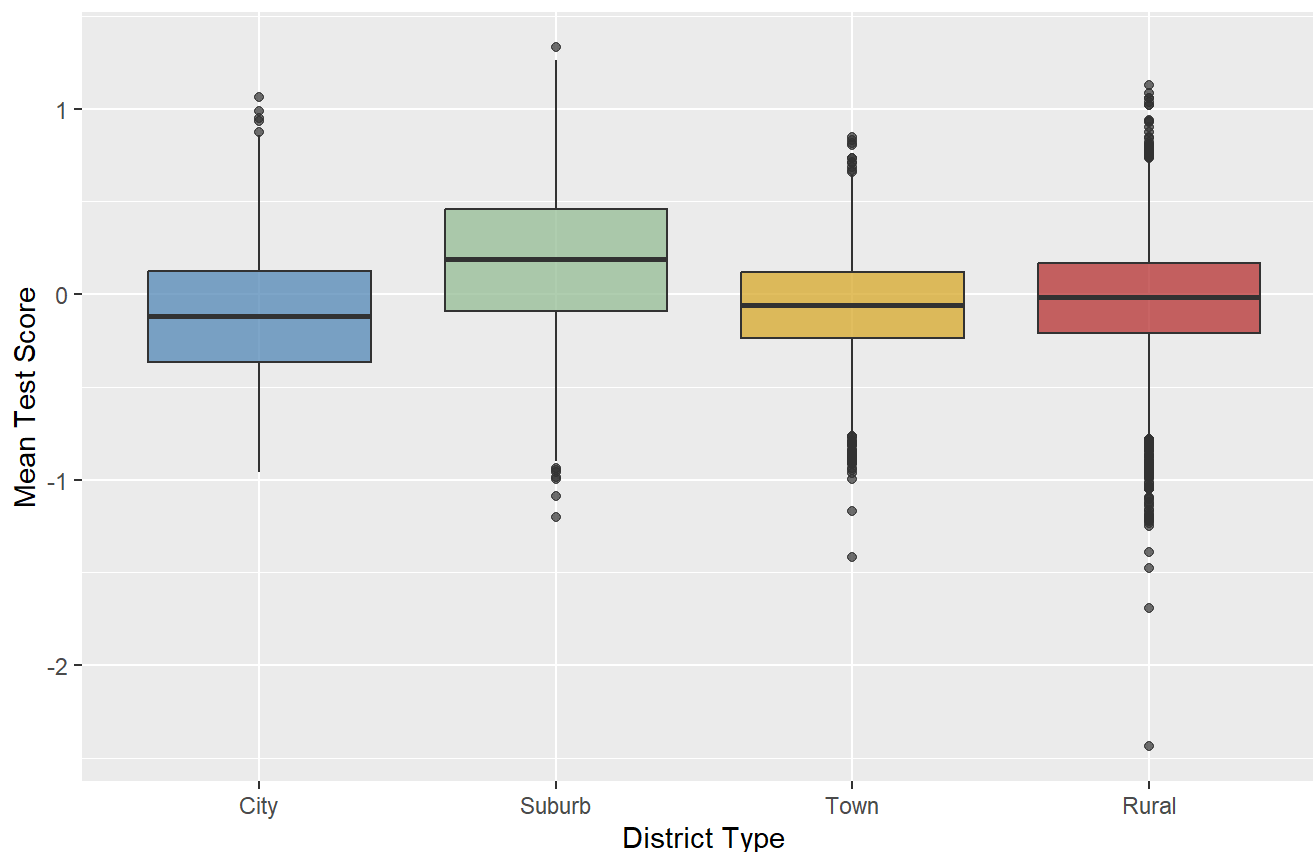
How Correlated are the two predictors



```
#Test scores by urbanicity
ggplot(df_all, aes(x = urbanicity, y = test_score, fill = urbanicity)) +
  geom_boxplot(alpha = 0.7) +
  scale_fill_manual(values = c("City" = "steelblue",
                                "Suburb" = "darkseagreen",
                                "Town" = "goldenrod",
                                "Rural" = "firebrick")) +
  labs(title = "Test Scores by Urbanicity",
        subtitle = "How does district type relate to academic performance?",
        x = "District Type",
        y = "Mean Test Score") +
  theme(legend.position = "none")
```

Test Scores by Urbanicity

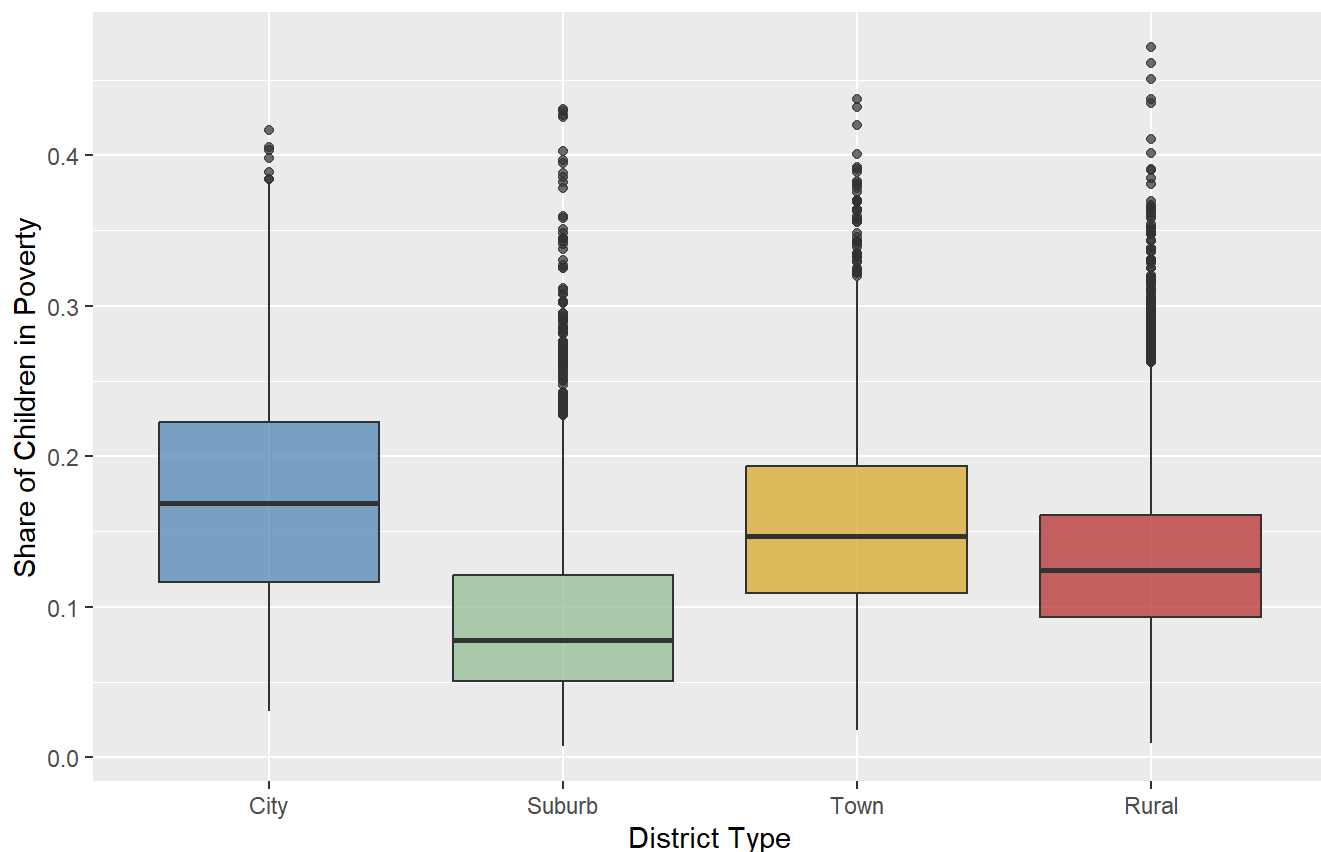
How does district type relate to academic performance?



```
#Neighborhood poverty by urbanicity
ggplot(df_all, aes(x = urbanicity, y = nbhd_poverty, fill = urbanicity)) +
  geom_boxplot(alpha = 0.7) +
  scale_fill_manual(values = c("City" = "steelblue",
                                "Suburb" = "darkseagreen",
                                "Town" = "goldenrod",
                                "Rural" = "firebrick")) +
  labs(title = "Neighborhood Poverty by Urbanicity",
        subtitle = "Do poverty rates differ across district types?",
        x = "District Type",
        y = "Share of Children in Poverty") +
  theme(legend.position = "none")
```

Neighborhood Poverty by Urbanicity

Do poverty rates differ across district types?



Frequentist Analysis

#Correlation plots - all students

```
cor.test(df_all$fam_income, df_all$test_score)
```

```
##
## Pearson's product-moment correlation
##
## data: df_all$fam_income and df_all$test_score
## t = 102.02, df = 12333, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.6668478 0.6859912
## sample estimates:
## cor
## 0.6765338
```

```
cor.test(df_all$nbhd_poverty, df_all$test_score)
```

```
##
## Pearson's product-moment correlation
##
## data: df_all$nbhd_poverty and df_all$test_score
## t = -104.09, df = 12333, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.6931358 -0.6743438
## sample estimates:
## cor
## -0.6838532
```

Regression Models

```
library(car)
```

```
## Warning: package 'car' was built under R version 4.4.2
```

```
## Loading required package: carData
```

```
## Warning: package 'carData' was built under R version 4.4.2
```

```
##
## Attaching package: 'car'
```

```
## The following object is masked from 'package:dplyr':
##
## recode
```

```
## The following object is masked from 'package:purrr':
##
## some
```

```
#All students
model1<- lm(test_score ~ fam_income + nbhd_poverty + urbanicity,data = df_all)

vif(model1)
```

```
##              GVIF Df GVIF^(1/(2*Df))
## fam_income   4.145344 1      2.036012
## nbhd_poverty 3.749433 1      1.936345
## urbanicity   1.441759 3      1.062875
```

```
summary(model1)
```

```
##
## Call:
## lm(formula = test_score ~ fam_income + nbhd_poverty + urbanicity,
##     data = df_all)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.72536 -0.14614  0.00496  0.15219  1.73169
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -3.649297    0.157007  -23.243 < 2e-16 ***
## fam_income      0.361982    0.013607   26.604 < 2e-16 ***
## nbhd_poverty   -2.136732    0.065020  -32.863 < 2e-16 ***
## urbanicitySuburb 0.005265    0.010565    0.498  0.61826
## urbanicityTown  0.030701    0.010643    2.884  0.00393 **
## urbanicityRural -0.005573    0.010062   -0.554  0.57967
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2391 on 12329 degrees of freedom
## Multiple R-squared:  0.5062, Adjusted R-squared:  0.506
## F-statistic: 2527 on 5 and 12329 DF, p-value: < 2.2e-16
```

Bayesian Method

```
#For all students
library(BayesFactor)
```

```
## Warning: package 'BayesFactor' was built under R version 4.4.3
```

```
## Loading required package: coda
```

```
## Warning: package 'coda' was built under R version 4.4.3
```

```
## Loading required package: Matrix
```

```
##
## Attaching package: 'Matrix'
```

```
## The following objects are masked from 'package:tidyr':
##
## expand, pack, unpack
```

```
## *****
## Welcome to BayesFactor 0.9.12-4.7. If you have questions, please contact Richard Morey (richardmorey@gmail.com).
##
## Type BFManual() to open the manual.
## *****
```

```
lmBF<- lmBF(test_score ~ fam_income + nbhd_poverty + urbanicity, data=df_all, posterior = TRUE,
iterations=10000)
```

```
## Warning: data coerced from tibble to data frame
```

```
summary(lmBF)
```

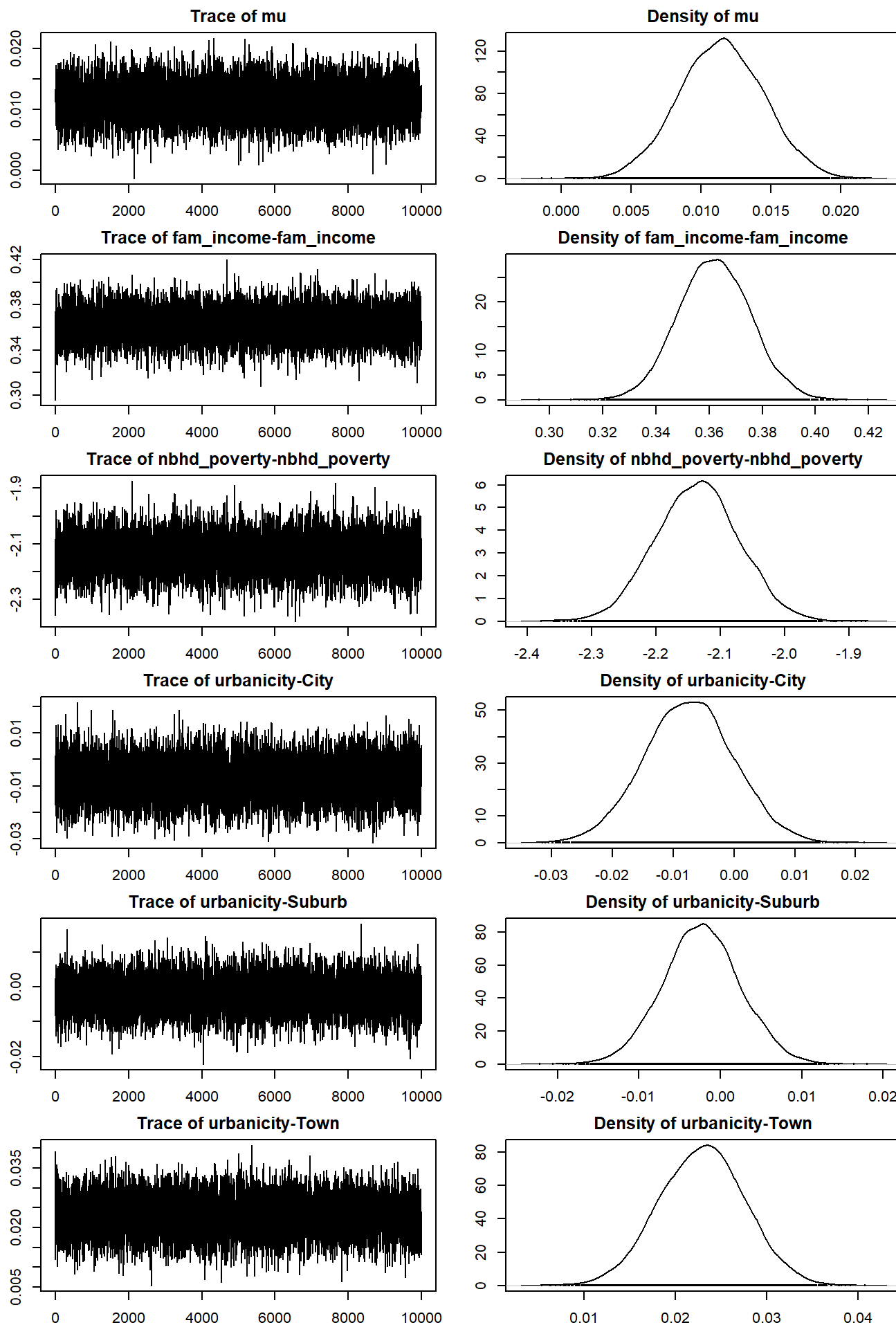
```
##
## Iterations = 1:10000
## Thinning interval = 1
## Number of chains = 1
## Sample size per chain = 10000
##
## 1. Empirical mean and standard deviation for each variable,
##    plus standard error of the mean:
##
##
```

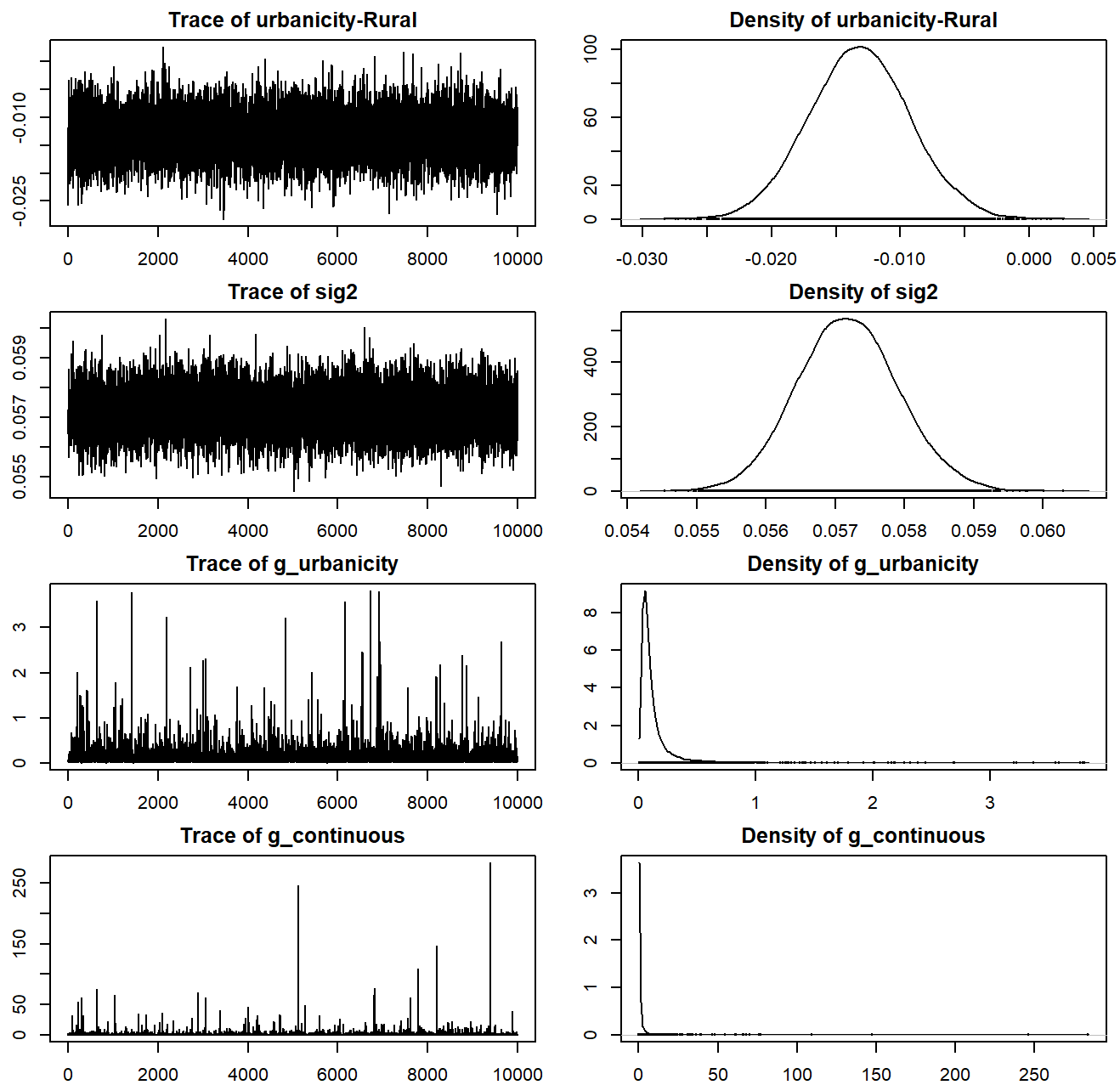
	Mean	SD	Naive SE	Time-series SE
## mu	0.011439	0.003007	3.007e-05	3.007e-05
## fam_income-fam_income	0.361897	0.013536	1.354e-04	1.354e-04
## nbhd_poverty-nbhd_poverty	-2.136404	0.064627	6.463e-04	6.463e-04
## urbanicity-City	-0.007394	0.007253	7.253e-05	7.253e-05
## urbanicity-Suburb	-0.002327	0.004742	4.742e-05	5.172e-05
## urbanicity-Town	0.022896	0.004649	4.649e-05	4.649e-05
## urbanicity-Rural	-0.013175	0.003903	3.903e-05	3.903e-05
## sig2	0.057184	0.000724	7.240e-06	7.240e-06
## g_urbanicity	0.125943	0.187306	1.873e-03	1.873e-03
## g_continuous	1.162762	5.136668	5.137e-02	5.137e-02

```
##
## 2. Quantiles for each variable:
##
##
```

	2.5%	25%	50%	75%	97.5%
## mu	0.005476	0.009399	0.011452	0.0135150	0.017375
## fam_income-fam_income	0.335671	0.352829	0.361923	0.3711142	0.388713
## nbhd_poverty-nbhd_poverty	-2.261602	-2.179657	-2.135793	-2.0939494	-2.010449
## urbanicity-City	-0.021494	-0.012282	-0.007406	-0.0025520	0.006883
## urbanicity-Suburb	-0.011580	-0.005457	-0.002324	0.0007913	0.007011
## urbanicity-Town	0.013752	0.019690	0.022974	0.0260482	0.031910
## urbanicity-Rural	-0.020774	-0.015810	-0.013190	-0.0105722	-0.005457
## sig2	0.055789	0.056691	0.057176	0.0576649	0.058642
## g_urbanicity	0.023420	0.048761	0.078395	0.1349117	0.526358
## g_continuous	0.122280	0.279915	0.489719	0.9516946	5.472914

```
par(mar = c(2, 2, 2, 2))  
plot(lmBF)
```





#Posterior Distribution Plots

```
post_all <- as.data.frame(lmBF)

colnames(post_all)
```

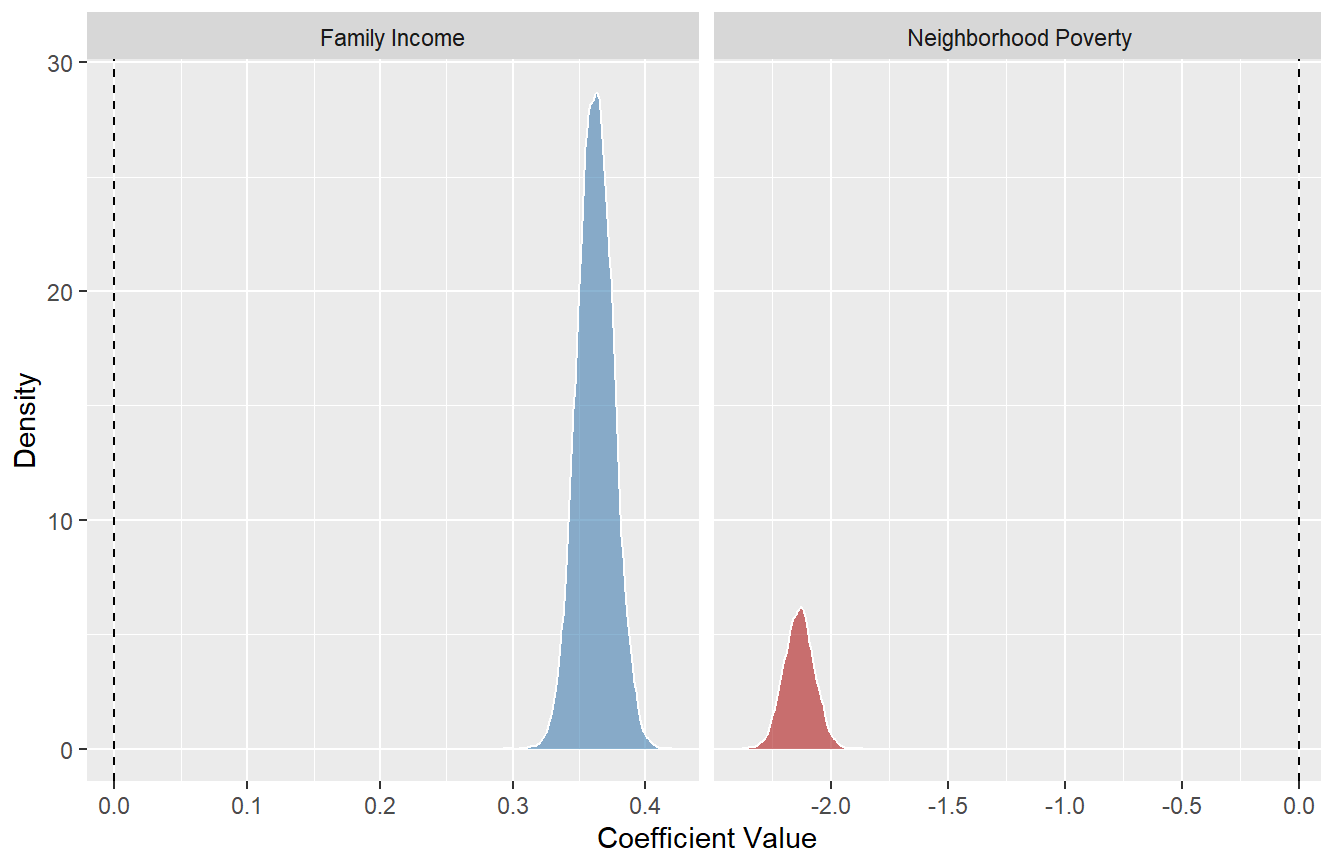
```
## [1] "mu" "fam_income-fam_income"
## [3] "nbhd_poverty-nbhd_poverty" "urbanicity-City"
## [5] "urbanicity-Suburb" "urbanicity-Town"
## [7] "urbanicity-Rural" "sig2"
## [9] "g_urbanicity" "g_continuous"
```

```
# — COMBINED POSTERIOR — family income and neighborhood poverty —————
post_long <- post_all %>%
  select(`fam_income-fam_income`, `nbhd_poverty-nbhd_poverty`) %>%
  pivot_longer(cols = everything(),
               names_to = "predictor",
               values_to = "value") %>%
  mutate(predictor = case_when(
    predictor == "fam_income-fam_income" ~ "Family Income",
    predictor == "nbhd_poverty-nbhd_poverty" ~ "Neighborhood Poverty"
  ))

ggplot(post_long, aes(x = value, fill = predictor)) +
  geom_density(alpha = 0.6, color = "white") +
  geom_vline(xintercept = 0, linetype = "dashed", color = "black") +
  scale_fill_manual(values = c("Family Income" = "steelblue",
                              "Neighborhood Poverty" = "firebrick")) +
  facet_wrap(~ predictor, scales = "free_x") +
  labs(title = "Posterior Distributions of Predictor Effects",
       subtitle = "Dashed line = zero effect",
       x = "Coefficient Value",
       y = "Density") +
  theme(legend.position = "none")
```

Posterior Distributions of Predictor Effects

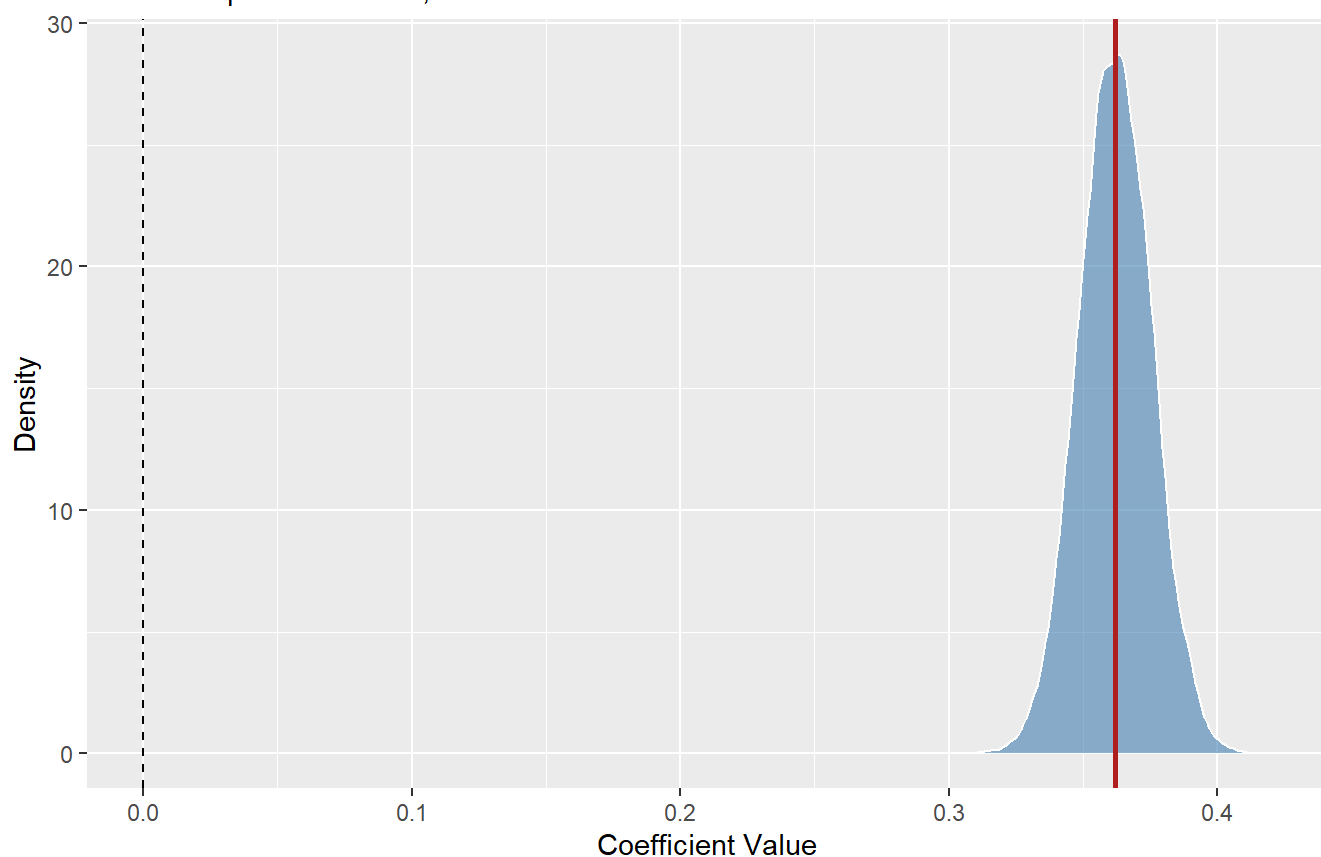
Dashed line = zero effect



```
# — FAMILY INCOME POSTERIOR —————
ggplot(post_all, aes(x = `fam_income-fam_income`)) +
  geom_density(fill = "steelblue", alpha = 0.6, color = "white") +
  geom_vline(xintercept = 0, linetype = "dashed", color = "black") +
  geom_vline(xintercept = mean(post_all$`fam_income-fam_income`),
    color = "firebrick", linewidth = 1) +
  labs(title = "Posterior Distribution – Family Income Effect",
    subtitle = "Red line = posterior mean, dashed = zero",
    x = "Coefficient Value",
    y = "Density")
```

Posterior Distribution — Family Income Effect

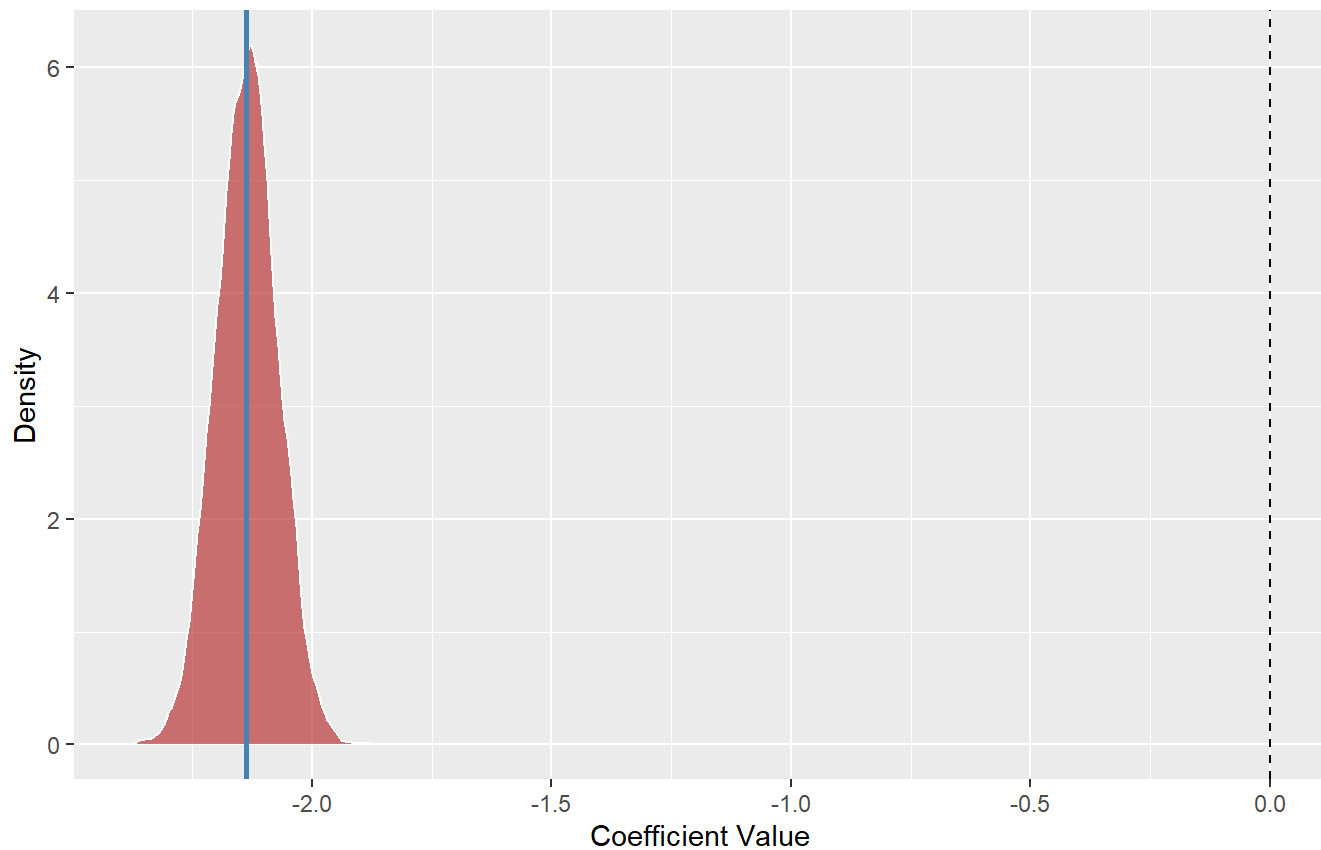
Red line = posterior mean, dashed = zero



```
# — NEIGHBORHOOD POVERTY POSTERIOR —————
ggplot(post_all, aes(x = `nbhd_poverty-nbhd_poverty`)) +
  geom_density(fill = "firebrick", alpha = 0.6, color = "white") +
  geom_vline(xintercept = 0, linetype = "dashed", color = "black") +
  geom_vline(xintercept = mean(post_all$`nbhd_poverty-nbhd_poverty`),
             color = "steelblue", linewidth = 1) +
  labs(title = "Posterior Distribution – Neighborhood Poverty Effect",
       subtitle = "Red line = posterior mean, dashed = zero",
       x = "Coefficient Value",
       y = "Density")
```

Posterior Distribution — Neighborhood Poverty Effect

Red line = posterior mean, dashed = zero



```
# — CREDIBLE INTERVALS —————
ci_all <- quantile(post_all$`nbhd_poverty-nbhd_poverty`,
  probs = c(0.025, 0.975))
```

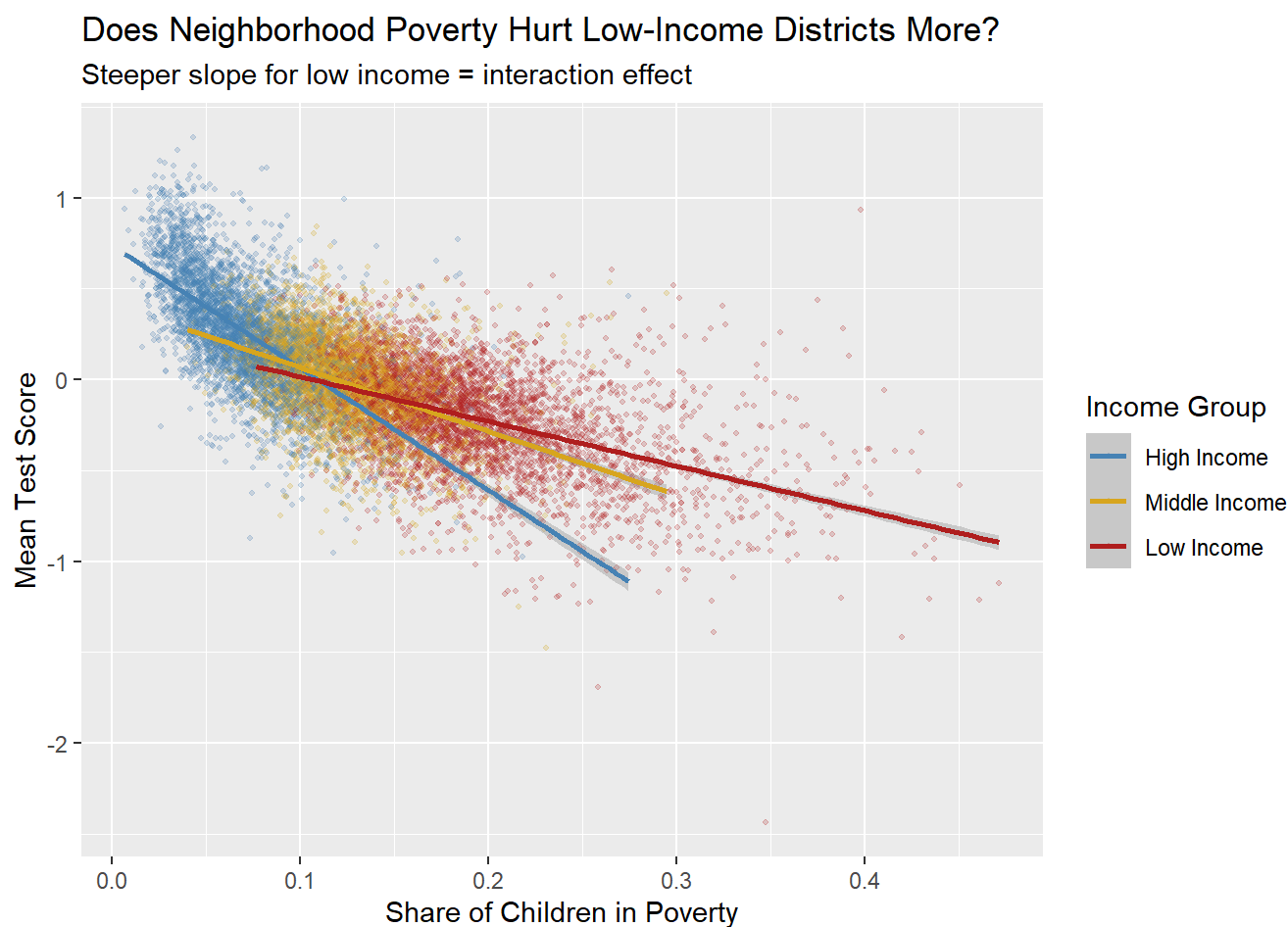
Supplemental Analysis about Urbanicity and

```
df_all <- df_all %>%
  mutate(income_group = case_when(
    fam_income < quantile(fam_income, 0.33) ~ "Low Income",
    fam_income < quantile(fam_income, 0.67) ~ "Middle Income",
    TRUE ~ "High Income"
  )) %>% factor(levels = c("High Income", "Middle Income", "Low Income"))
```

Interaction Plots

```
#Income group and neighborhood poverty
ggplot(df_all, aes(x = nbhd_poverty, y = test_score, color = income_group)) +
  geom_point(alpha = 0.2, size = 0.7) +
  geom_smooth(method = "lm", se = TRUE) +
  scale_color_manual(values = c("Low Income" = "firebrick",
                                "Middle Income" = "goldenrod",
                                "High Income" = "steelblue")) +
  labs(title = "Does Neighborhood Poverty Hurt Low-Income Districts More?",
       subtitle = "Steeper slope for low income = interaction effect",
       x = "Share of Children in Poverty",
       y = "Mean Test Score",
       color = "Income Group")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



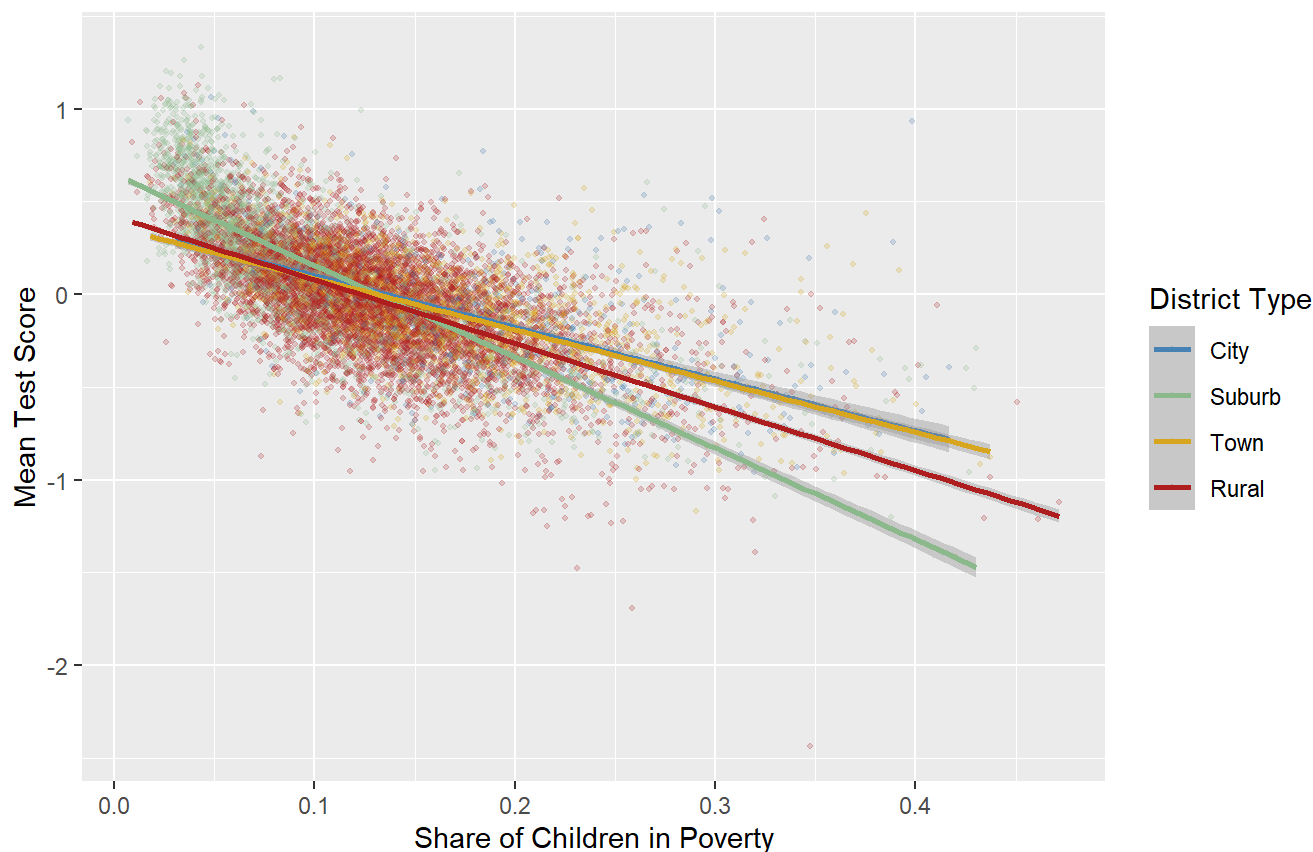
```
#Urbanicity and neighborhood poverty
```

```
ggplot(df_all, aes(x = nbhd_poverty, y = test_score, color = urbanicity)) +  
  geom_point(alpha = 0.2, size = 0.7) +  
  geom_smooth(method = "lm", se = TRUE) +  
  scale_color_manual(values = c("City" = "steelblue",  
                                "Suburb" = "darkseagreen",  
                                "Town" = "goldenrod",  
                                "Rural" = "firebrick")) +  
  labs(title = "Neighborhood Poverty vs. Test Scores by District Type",  
        subtitle = "Different slopes = interaction effect",  
        x = "Share of Children in Poverty",  
        y = "Mean Test Score",  
        color = "District Type")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

Neighborhood Poverty vs. Test Scores by District Type

Different slopes = interaction effect



Frequentist Interaction Models

```
# Interaction 1: poverty and income group  
model2<- lm(test_score ~ fam_income + nbhd_poverty * income_group,data = df_all)  
summary(model2)
```

```
##
## Call:
## lm(formula = test_score ~ fam_income + nbhd_poverty * income_group,
##     data = df_all)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.81482 -0.14037  0.00212  0.14628  1.63048
##
## Coefficients:
##                                Estimate Std. Error t value Pr(>|t|)
## (Intercept)                   -3.54717     0.19783  -17.930 < 2e-16 ***
## fam_income                     0.36991     0.01706   21.687 < 2e-16 ***
## nbhd_poverty                  -4.85402     0.15120  -32.104 < 2e-16 ***
## income_groupMiddle Income    -0.08344     0.02094   -3.986 6.77e-05 ***
## income_groupLow Income      -0.21449     0.02033  -10.551 < 2e-16 ***
## nbhd_poverty:income_groupMiddle Income  1.72373     0.18465    9.335 < 2e-16 ***
## nbhd_poverty:income_groupLow Income    3.04993     0.15120   20.172 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2329 on 12328 degrees of freedom
## Multiple R-squared:  0.5316, Adjusted R-squared:  0.5314
## F-statistic: 2332 on 6 and 12328 DF, p-value: < 2.2e-16
```

```
# Interaction 2: poverty and urbanicity
```

```
freq_int_urban <- lm(test_score ~ fam_income + nbhd_poverty * urbanicity, data = df_all)
summary(freq_int_urban)
```

```
##
## Call:
## lm(formula = test_score ~ fam_income + nbhd_poverty * urbanicity,
##     data = df_all)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.74494 -0.14383  0.00383  0.14899  1.59310
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -3.20887     0.15895  -20.187 < 2e-16 ***
## fam_income       0.31340     0.01375   22.794 < 2e-16 ***
## nbhd_poverty    -1.64619     0.12744  -12.917 < 2e-16 ***
## urbanicitySuburb  0.22527     0.02382    9.458 < 2e-16 ***
## urbanicityTown   0.05028     0.02589    1.942  0.05214 .
## urbanicityRural  0.08471     0.02359    3.591  0.00033 ***
## nbhd_poverty:urbanicitySuburb -1.75439     0.13899  -12.622 < 2e-16 ***
## nbhd_poverty:urbanicityTown  -0.09999     0.13983   -0.715  0.47459
## nbhd_poverty:urbanicityRural -0.53828     0.12878   -4.180 2.94e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2363 on 12326 degrees of freedom
## Multiple R-squared:  0.5181, Adjusted R-squared:  0.5178
## F-statistic: 1657 on 8 and 12326 DF,  p-value: < 2.2e-16
```

Bayesian Interaction Models

```
# Interaction 1: poverty and income group
lmBF2 <- lmBF(
  test_score ~ fam_income + nbhd_poverty + income_group + nbhd_poverty:income_group,
  data      = as.data.frame(df_all),
  posterior = TRUE,
  iterations = 10000
)
summary(lmBF2)
```

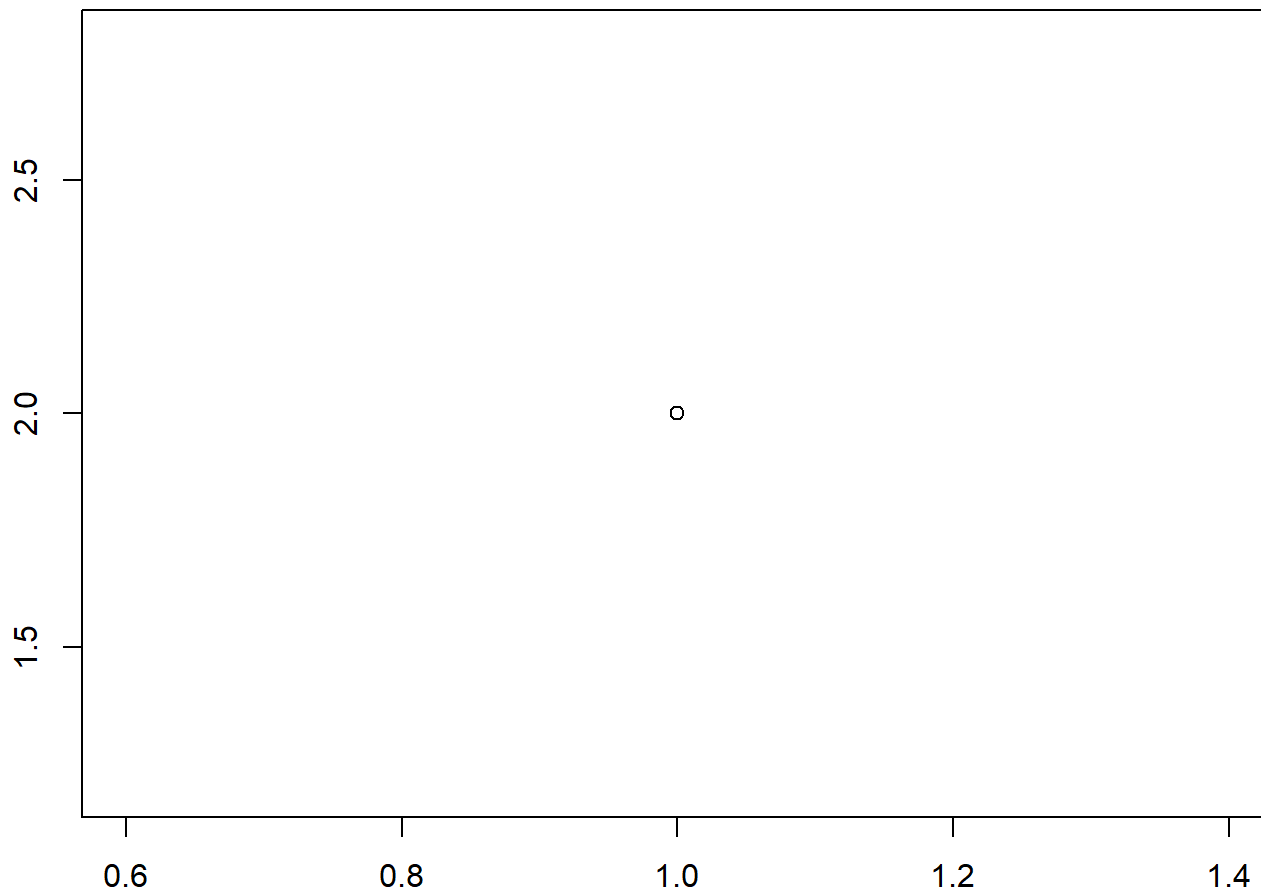
```

##
## Iterations = 1:10000
## Thinning interval = 1
## Number of chains = 1
## Sample size per chain = 10000
##
## 1. Empirical mean and standard deviation for each variable,
##    plus standard error of the mean:
##
##              Mean          SD  Naive SE
## mu              -0.05347 0.0036379 3.638e-05
## fam_income-fam_income      0.36887 0.0169817 1.698e-04
## nbhd_poverty-nbhd_poverty  -3.25805 0.0754460 7.545e-04
## income_group-High Income   -0.10650 0.0062473 6.247e-05
## income_group-Middle Income  0.03286 0.0040892 4.089e-05
## income_group-Low Income     0.07364 0.0053562 5.356e-05
## nbhd_poverty:income_group-High Income -1.58378 0.1035789 1.036e-03
## nbhd_poverty:income_group-Middle Income 0.12801 0.0959887 9.599e-04
## nbhd_poverty:income_group-Low Income   1.45577 0.0730365 7.304e-04
## sig2              0.05425 0.0006925 6.925e-06
## g_income_group      0.61181 6.1847959 6.185e-02
## g_continuous        0.64691 0.7315807 7.316e-03
##
##              Time-series SE
## mu              3.638e-05
## fam_income-fam_income      1.698e-04
## nbhd_poverty-nbhd_poverty  7.545e-04
## income_group-High Income   6.247e-05
## income_group-Middle Income  4.075e-05
## income_group-Low Income     5.451e-05
## nbhd_poverty:income_group-High Income  1.036e-03
## nbhd_poverty:income_group-Middle Income 9.730e-04
## nbhd_poverty:income_group-Low Income   7.304e-04
## sig2              6.925e-06
## g_income_group      6.185e-02
## g_continuous        7.573e-03
##
## 2. Quantiles for each variable:
##
##              2.5%      25%      50%      75%
## mu              -0.06087 -0.05589 -0.05342 -0.05099
## fam_income-fam_income      0.33501 0.35738 0.36890 0.38022
## nbhd_poverty-nbhd_poverty  -3.40736 -3.30724 -3.25839 -3.20689
## income_group-High Income   -0.11877 -0.11071 -0.10650 -0.10227
## income_group-Middle Income  0.02492 0.03011 0.03287 0.03561
## income_group-Low Income     0.06319 0.07009 0.07367 0.07731
## nbhd_poverty:income_group-High Income -1.78897 -1.65253 -1.58371 -1.51498
## nbhd_poverty:income_group-Middle Income -0.05973 0.06309 0.12677 0.19247
## nbhd_poverty:income_group-Low Income   1.31464 1.40644 1.45556 1.50563
## sig2              0.05292 0.05379 0.05425 0.05472
## g_income_group      0.06195 0.13961 0.24463 0.47215
## g_continuous        0.14911 0.29364 0.44685 0.73353
##
##              97.5%

```

```
## mu -0.04636
## fam_income-fam_income 0.40131
## nbhd_poverty-nbhd_poverty -3.11168
## income_group-High Income -0.09429
## income_group-Middle Income 0.04087
## income_group-Low Income 0.08386
## nbhd_poverty:income_group-High Income -1.38075
## nbhd_poverty:income_group-Middle Income 0.32088
## nbhd_poverty:income_group-Low Income 1.60034
## sig2 0.05563
## g_income_group 2.67601
## g_continuous 2.39845
```

```
par(mar = c(2, 2, 2, 2))
plot(2)
```



```
lmBF3 <- lmBF(
  test_score ~ fam_income + nbhd_poverty + urbanicity + nbhd_poverty:urbanicity,
  data       = as.data.frame(df_all),
  posterior  = TRUE,
  iterations = 10000
)
summary(lmBF3)
```

```

##
## Iterations = 1:10000
## Thinning interval = 1
## Number of chains = 1
## Sample size per chain = 10000
##
## 1. Empirical mean and standard deviation for each variable,
##    plus standard error of the mean:
##
##              Mean          SD Naive SE Time-series SE
## mu          -0.007164 0.0032682 3.268e-05      3.268e-05
## fam_income-fam_income      0.312740 0.0136914 1.369e-04      1.390e-04
## nbhd_poverty-nbhd_poverty -2.244946 0.0679418 6.794e-04      6.794e-04
## urbanicity-City          -0.012018 0.0081485 8.148e-05      8.148e-05
## urbanicity-Suburb        -0.014896 0.0049941 4.994e-05      4.988e-05
## urbanicity-Town          0.024634 0.0051307 5.131e-05      5.131e-05
## urbanicity-Rural         0.002279 0.0039929 3.993e-05      3.993e-05
## nbhd_poverty:urbanicity-City 0.596794 0.0925717 9.257e-04      9.257e-04
## nbhd_poverty:urbanicity-Suburb -1.154773 0.0678693 6.787e-04      6.602e-04
## nbhd_poverty:urbanicity-Town 0.498294 0.0684103 6.841e-04      6.856e-04
## nbhd_poverty:urbanicity-Rural 0.059685 0.0561024 5.610e-04      5.610e-04
## sig2              0.055812 0.0007046 7.046e-06      7.273e-06
## g_urbanicity          0.151102 1.8307762 1.831e-02      1.831e-02
## g_continuous          0.308486 0.2870565 2.871e-03      2.871e-03
##
## 2. Quantiles for each variable:
##
##              2.5%          25%          50%          75%
## mu          -0.013587 -0.0094027 -0.007127 -0.004998
## fam_income-fam_income      0.285718 0.3034407 0.312638 0.322121
## nbhd_poverty-nbhd_poverty -2.377103 -2.2905859 -2.245202 -2.199157
## urbanicity-City          -0.028007 -0.0173825 -0.012069 -0.006549
## urbanicity-Suburb        -0.024590 -0.0182835 -0.014894 -0.011536
## urbanicity-Town          0.014599 0.0211061 0.024692 0.028104
## urbanicity-Rural         -0.005654 -0.0004046 0.002262 0.004926
## nbhd_poverty:urbanicity-City 0.417045 0.5340657 0.598317 0.659737
## nbhd_poverty:urbanicity-Suburb -1.285232 -1.2016596 -1.154805 -1.108778
## nbhd_poverty:urbanicity-Town 0.364784 0.4520332 0.498293 0.543735
## nbhd_poverty:urbanicity-Rural -0.047886 0.0215657 0.060080 0.096479
## sig2              0.054427 0.0553506 0.055800 0.056282
## g_urbanicity          0.024289 0.0500299 0.080903 0.142142
## g_continuous          0.086336 0.1585530 0.230508 0.356186
##
##              97.5%
## mu          -0.0006874
## fam_income-fam_income      0.3395198
## nbhd_poverty-nbhd_poverty -2.1127023
## urbanicity-City          0.0041157
## urbanicity-Suburb        -0.0051600
## urbanicity-Town          0.0346583
## urbanicity-Rural         0.0102028
## nbhd_poverty:urbanicity-City 0.7750624
## nbhd_poverty:urbanicity-Suburb -1.0220780

```

```
## nbhd_poverty:urbanicity-Town    0.6334829
## nbhd_poverty:urbanicity-Rural   0.1708225
## sig2                             0.0572006
## g_urbanicity                    0.5668746
## g_continuous                     1.0122207
```

```
par(mar = c(2, 2, 2, 2))
plot(lmBF3)
```

